A Mini Project Report (Data Analysis) On

PYTHON MINI PROJECT

Submitted in partial fulfilment for the award of degree of

B.C.A. (Honors)

in

School of Computer Science and Engineering

CS1280 Python Programming for Data Analytics

Submitted by

VAISHNAVI .P.M 1RVUA25BCA0114

School of Computer Science and Engineering

RV University, Bengaluru - 560059

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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

C E R T I F I C A T E

This is to be certified that the project work titled **PYTHON MINI PROJECT** is carried out by VAISHNAVI .P.M , 1RVUA25BCA0114 a Bonafide student of RV University, Bengaluru, in partial fulfilment of Bachelor of Computer Applications (Hons) in School of Computer Science and Engineering of the RV University, Bengaluru during the year 2025-2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the mini project report.

Dr. Mahadev Patil / Dr. Thenmozhi K.
Subject Teacher

D E C L A R A T I O N

I, **Vaishnavi pm (1RVUA25BCA0114)** student of 2nd semester BCA **,** at School of Computer Science and Engineering, **RV University,** hereby declare that the project work titled "**PYTHON MINI PROJECT**" has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Computer Application** during the academic year **2026-2027**. Further, the matter presented in the project has not been submitted previously by anybody for the award of any degree or any diploma to any other University, to the best of our knowledge and faith.

Name: Vaishnavi.P.M USN: 1RUA25BCA0114

Place: Bengaluru -560073

Date: -04-2026 Signature of student

Name of the Faculty: Signature with date of Faculty

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I N T R O D U C T I O N

Data analytics involves collecting, cleaning, and analyzing data to extract meaningful insights. In this project, we perform end-to-end data analysis using Python on a real-world dataset. The process includes data preprocessing, handling missing values, detecting outliers, and visualizing data to understand patterns and trends.

O B J E C T I V E S

- To load and explore the dataset
- To clean data (remove duplicates, handle missing values)
- To detect outliers and preprocess data
- To encode categorical variables and scale features
- To analyze data using grouping and aggregation
- To visualize data and derive meaningful insights

C O D E S T R U C T U R E

1.LOAD DATA:

We import the dataset into a Pandas DataFrame using read_csv. This step makes the raw data accessible for analysis in Python.

```
>>> #LOAD DATA
>>> import pandas as pd
>>> df = pd.read_csv("C:/Users/Vaishnavi PM/OneDrive/Desktop/india_colleges.csv")
>>>
```

2.CHECK SHAPE:

We inspect the number of rows and columns with .shape and .info(). This helps confirm dataset size and understand data types.

```
>>> #CHECK SHAPE
>>> print(df.shape)
(1203, 8)
>>> print(df.info())
<class 'pandas.DataFrame'>
RangeIndex: 1203 entries, 0 to 1202
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   name             1203 non-null   str
 1   city             733 non-null    str
 2   state            1203 non-null   str
 3   type             1203 non-null   str
 4   fees_ug_inr      1203 non-null   float64
 5   placement_avg_lpa  1203 non-null   float64
 6   rating           1203 non-null   float64
 7   nirf_rank        207 non-null    float64
dtypes: float64(4), str(4)
memory usage: 75.3 KB
None
>>>
>>>
```

3.REMOVE DUPLICATES:

Duplicate rows are dropped using .drop_duplicates(). This ensures each record is unique and prevents bias in analysis.

```
>>> #REMOVE DUPLICATES
>>> df = df.drop_duplicates()
>>> print(df.shape)
    (1203, 8)
>>> #NO DUPLICATES
>>>
>>>
```

4.HANDLE MISSING VALUES:

Missing values are filled with mean (numeric) or mode (categorical). This prevents errors in calculations and keeps the dataset consistent.

```
>>> #HANDLE MISSING VALUES
>>> # Numeric → mean
>>> num_cols = df.select_dtypes(include=['number']).columns
>>> df[num_cols] = df[num_cols].fillna(df[num_cols].mean())
>>> # String → mode
>>> str_cols = df.select_dtypes(include=['str']).columns
>>> for col in str_cols:
...     df[col] = df[col].fillna(df[col].mode()[0])
...
...
>>>
```

5.DETECT OUTLIERS:

Outliers are identified using boxplots or statistical methods (IQR/Z-score). They highlight extreme values that may distort analysis

```
>>> #DETECT OUTLIERS
>>> import seaborn as sns
>>> #horizontal boxplot
>>> sns.boxplot(x=df["fees_ug_inr"])
    <Axes: xlabel='fees_ug_inr'>
>>> Q1 = df["fees_ug_inr"].quantile(0.25)
>>> Q3 = df["fees_ug_inr"].quantile(0.75)
>>> IQR = Q3 - Q1
>>> outliers = df[(df["fees_ug_inr"] < (Q1 - 1.5*IQR)) | (df["fees_ug_inr"] > (Q3 + 1.5*IQR))]
>>> print(outliers[["name","fees_ug_inr"]].head())
            name  fees_ug_inr
    74  IIM Ahmedabad    2630000.0
    75   IIM Amritsar     965000.0
    76  IIM Bangalore    2630000.0
    77  IIM Bodh Gaya    1400000.0
    78   IIM Calcutta    2500000.0
>>>
```

6.ENCODE CATEGORICAL:

Categorical columns are converted into numeric dummy variables with get_dummies(). This makes them usable in models and comparisons.

```
>>> #ENCODE CATEGORICAL
>>> df = pd.get_dummies(df, columns=["type"])
>>>
>>> df.sample(5)
                                        name  ... type_University
    92                        IIM Tiruchirappalli  ...          False
    423                       Galgotias University  ...          False
    1192                  World University of Design  ...          False
    369              DJ Sanghvi College of Engineering  ...          False
    1165  Velagapudi Ramakrishna Siddhartha Engineering ...  ...          False

    [5 rows x 21 columns]
>>>
>>>
```

7.SCALE FEATURES:

Numeric features are standardized or normalized using scalers. This ensures all features are on a comparable scale for fair analysis.

```
>>> #SCALE FEATURES
>>> from sklearn.preprocessing import StandardScaler
>>> scaler = StandardScaler()
>>> scaled = scaler.fit_transform(df[["fees_ug_inr","placement_avg_lpa","rating","nirf_rank"]])
>>> df_scaled = pd.DataFrame(scaled, columns=num_cols)
>>> print(df_scaled.head())
   fees_ug_inr  placement_avg_lpa    rating  nirf_rank
0    -0.026004           0.204761  1.138799  -2.384958
1    -0.036305           0.204761  0.065057   4.153933
2    -0.036305           1.150579  1.377409  -2.793638
3    -0.028064           0.441216  0.542276  -0.423290
4    -0.019824           1.104470  1.735323  -3.365791
>>>
>>>
```

8.GROUP & AGGREGATE:

Data is grouped by categories (e.g., state, type) and aggregated with mean or sum. This reveals trends and comparisons across groups.

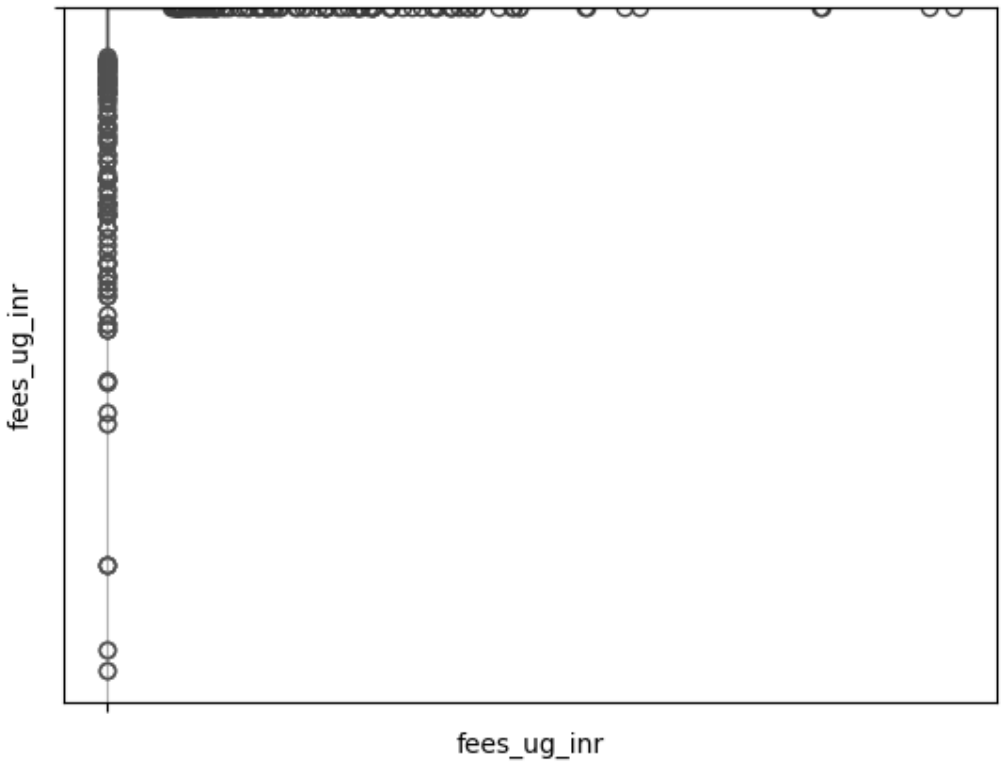
```
>>> #GROUP & AGGREGATE
>>> df.groupby("state")["placement_avg_lpa"].mean().head()
state
Andaman             6.000000
Andhra Pradesh      8.096486
Arunachal Pradesh   6.885714
Assam               6.960000
Bihar               7.294727
Name: placement_avg_lpa, dtype: float64
>>>
>>>
```

9. VISUALIZE:

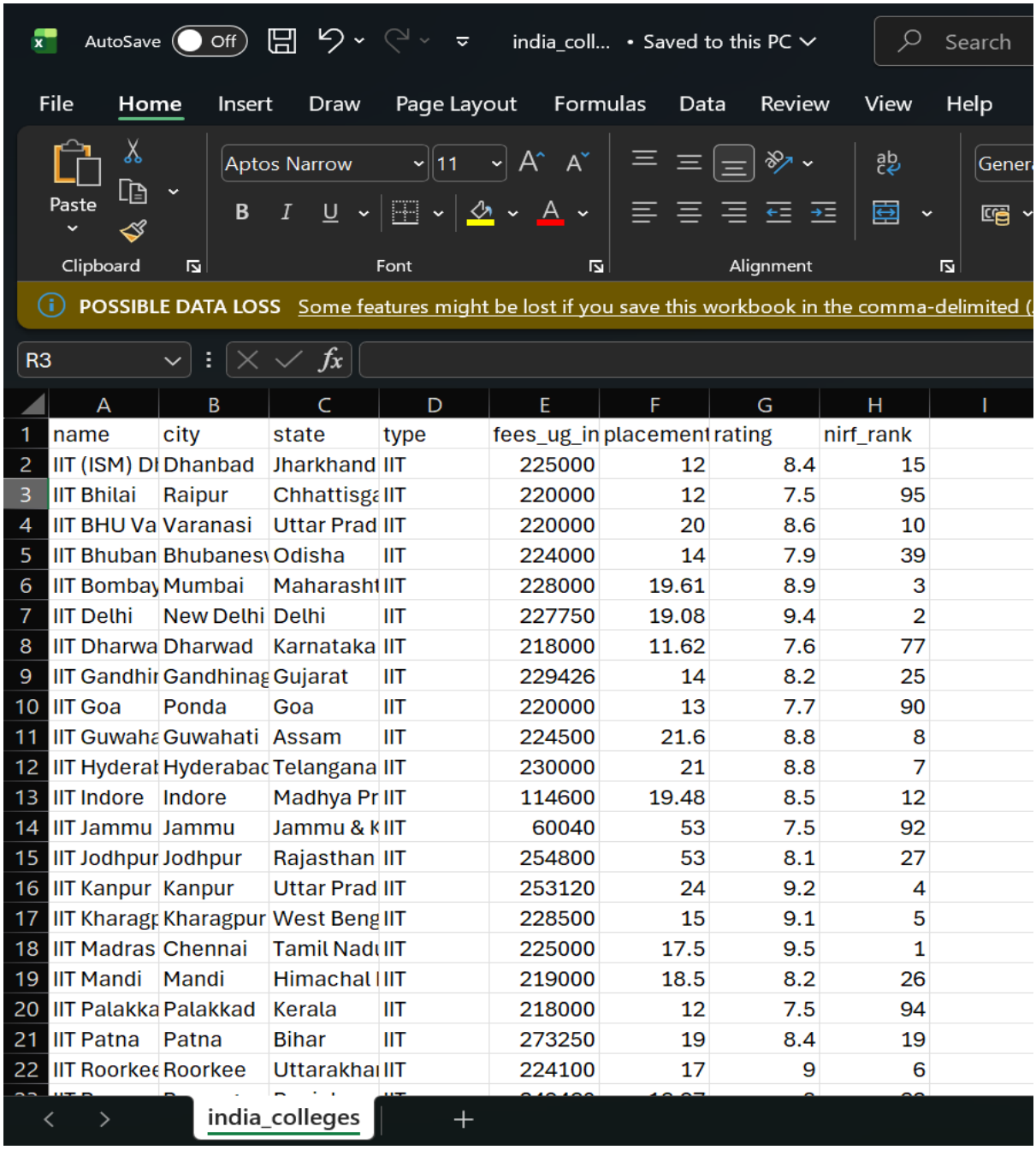
Charts like histograms, bar plots, and scatter plots are created. Visualization makes patterns, distributions, and insights easy to interpret.

```
>>> #VISUALIZE
>>> import matplotlib.pyplot as plt
>>> df["fees_ug_inr"].hist(bins=30)
    <Axes: xlabel='fees_ug_inr', ylabel='fees_ug_inr'>
>>> plt.show()

...
...
```



DATASET USED:



	A	B	C	D	E	F	G	H
1	name	city	state	type	fees_ug_in	placement	rating	nirf_rank
2	IIT (ISM) DI	Dhanbad	Jharkhand	IIT	225000	12	8.4	15
3	IIT Bhilai	Raipur	Chhattisga	IIT	220000	12	7.5	95
4	IIT BHU Va	Varanasi	Uttar Prad	IIT	220000	20	8.6	10
5	IIT Bhuban	Bhubanesv	Odisha	IIT	224000	14	7.9	39
6	IIT Bombay	Mumbai	Maharasht	IIT	228000	19.61	8.9	3
7	IIT Delhi	New Delhi	Delhi	IIT	227750	19.08	9.4	2
8	IIT Dharwa	Dharwad	Karnataka	IIT	218000	11.62	7.6	77
9	IIT Gandhir	Gandhinag	Gujarat	IIT	229426	14	8.2	25
10	IIT Goa	Ponda	Goa	IIT	220000	13	7.7	90
11	IIT Guwaha	Guwahati	Assam	IIT	224500	21.6	8.8	8
12	IIT Hyderat	Hyderabac	Telangana	IIT	230000	21	8.8	7
13	IIT Indore	Indore	Madhya Pr	IIT	114600	19.48	8.5	12
14	IIT Jammu	Jammu	Jammu & K	IIT	60040	53	7.5	92
15	IIT Jodhpur	Jodhpur	Rajasthan	IIT	254800	53	8.1	27
16	IIT Kanpur	Kanpur	Uttar Prad	IIT	253120	24	9.2	4
17	IIT Kharagp	Kharagpur	West Beng	IIT	228500	15	9.1	5
18	IIT Madras	Chennai	Tamil Nadu	IIT	225000	17.5	9.5	1
19	IIT Mandi	Mandi	Himachal I	IIT	219000	18.5	8.2	26
20	IIT Palakka	Palakkad	Kerala	IIT	218000	12	7.5	94
21	IIT Patna	Patna	Bihar	IIT	273250	19	8.4	19
22	IIT Roorkee	Roorkee	Uttarakhai	IIT	224100	17	9	6

LINK FOR THE DATASET:

https://www.kaggle.com/datasets/nagarshivam/india-colleges-and-cities

O U T C O M E

The project successfully performs end-to-end data analysis on a real-world dataset using Python. Cleaned and processed data is analyzed to identify patterns and trends. Visualizations help in understanding the data better, and meaningful insights are derived to support decision-making.

C O N C L U S I O N

Through this end-to-end data analysis, we successfully cleaned, transformed, and explored the India Colleges dataset. Each preprocessing step — from handling missing values and encoding categorical data to scaling features and detecting outliers — ensured the dataset was reliable and ready for meaningful insights. The analysis highlighted clear trends, such as significant fee variations across institution types and strong placement outcomes in premier institutes, while visualizations made these patterns easy to interpret. Overall, the project demonstrates how structured data analysis can uncover valuable comparisons and support informed decision-making.

….THANK YOU ….